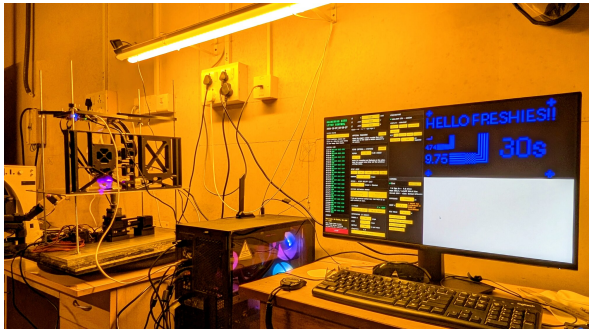
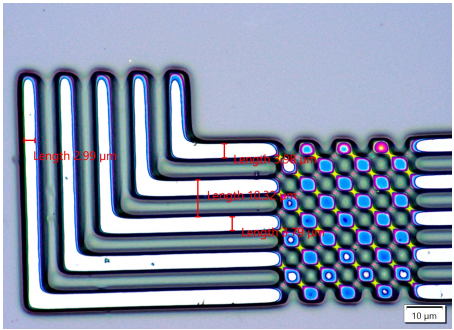
HackerFab: India's first student-built fab



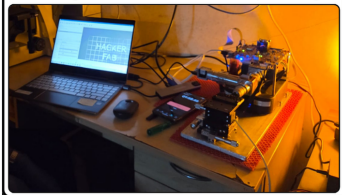
Direct-write photolithography setup

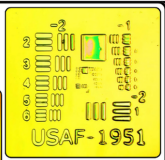


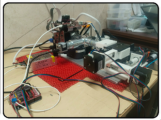
3 µm lines patterned by our system

DLP-BASED MASKLESS LITHOGRAPHY

- Patterns of resolution 10um with our DLP based maskless lithography system
- XYZ stage allows for automated multi-patterning







TUBE FURNACE

- Quartz-based tube furnace, capable of reaching 1200 C
- Built for oxidation as well as a blueprint for future CVD systems

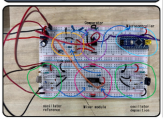




QC MONITOR

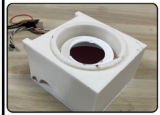
Quartz-crystal monitor built alongside the sputter system to measure deposition rate

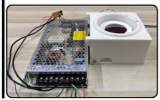




SPIN COATER

Spin coats wafers at upto 6000 rpm with 10 rpm error control for ramp speed

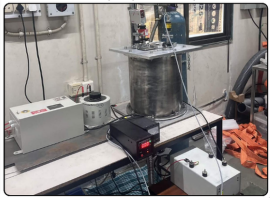






DC PLASMA SPUTTER

- Argon plasma sputtering for film deposition
- Magnetron system for directing particles
- Turbomolecular pump to reach upto 1e-5 mbar



HackerFab IITB

democratizing semiconductor fabrication

hackerfabiitb.github.io
hackerfabiitb@gmail.com

1. **Who we are:** A student-led R&D project guided by IIT Bombay SemiX faculty. We are building low-cost open-source student-accessible tools for making chips and transistors.
2. **Our Mission:** To develop a low-cost DIY semiconductor fab capable of making working transistors, whose open-source recipe can be replicated and developed by colleges across India.
3. **Our Vision:** To empower every EE student in India to learn to fabricate semiconductor devices (like a Tinkerer's Lab or Makerspace for semiconductors)
4. **Advised by SemiX faculty:** Prof. Swaroop Ganguly, Prof. Udayan Ganguly, Prof. Anil Kottantharayil and Prof. Veeresh Deshpande
5. **Recognition:**
   - 5.1 won User Design Track at 39th International Conf. On VLSI Design for our DIY toolkit & **Fab-in-a-Box** proposal
   - 5.2 Interview and article by Swarajya Magazine[1]
   - 5.3 Deep-dive video by RuntimeBRT[2]



Team at SEMICON India 2026

FROM WAFER TO TRANSISTOR: OUR TOOLCHAIN

One in-house tool for each front-end process step

OXIDISE
Tube furnace
1100 °C, single zone
~₹35k build

→

PATTERN
DMD direct-write litho
405 nm, 10× objective
< ₹2 lakh

→

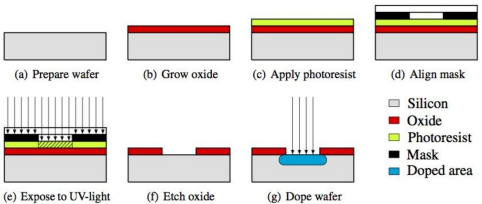
ETCH
HF / BOE wet etch
Oxide through PPR mask
Wet bench + fume hood

→

DOPE
Spin-on dopant
Furnace drive-in
No ion implanter

→

METALLISE
DC magnetron sputter
Thermal evaporator
Same vacuum chamber



TODAY'S ROUTE

- Lithography: how we built a DMD stepper
- Furnace: design, sizing, Deal–Grove
- Sputter, evaporator, QCM, RF plan
- Doping, MOSCAP and NMOS flows

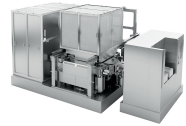
Design and Development of a $<$\$2000 Direct-Write Photolithography System

Range of Lithography Systems

Sr No	Tool Name	Resolution (nm)	Throughput (wafers/hr)	Cost (USD)
1	ASML TWINSCAN EXE:5200B High-NA EUV scanner[3, 4]	8	175	$380M
2	IMS MBMW-101 / -301 multi-beam mask writer[5, 6, 7]	14	0.1 *masks*/hr	$80M
3	Raith VOYAGER 50 kV e-beam lithography[8]	8	0.02	$5M
4	Durham Magneto Optics MicroWriter ML3 Pro maskless laser direct-write[9, 10]	600	0.4	$0.5M
5	**HackerFab IITB Litho** DMD projection direct-write	3000	0.003	**$2k**



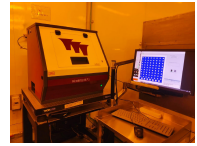
1. ASML EXE:5200B



2. IMS MBMW-101



3. Raith VOYAGER

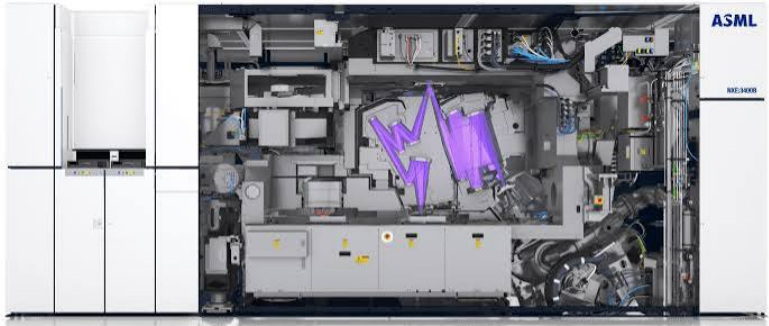


4. MicroWriter ML3



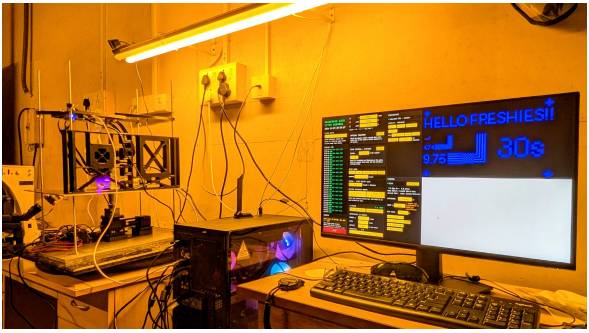
5. HackerFab Litho

ASML EUV

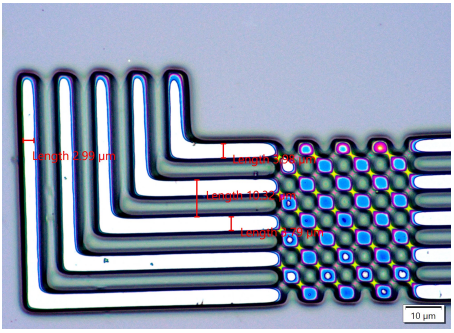


ASML NXE:3400B EUV scanner (CNBC)[11]

HackerFab Litho

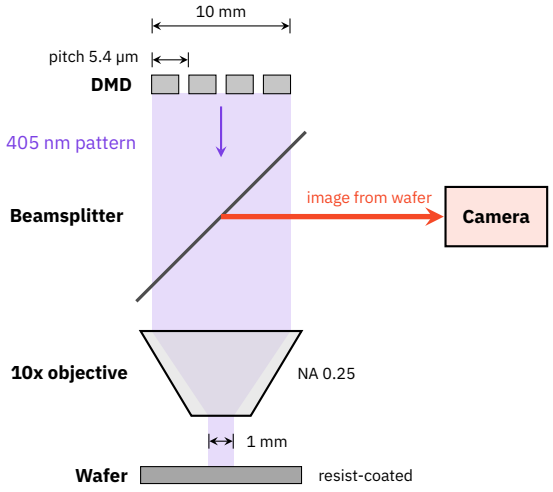


Direct-write photolithography setup



3 µm lines patterned by our system

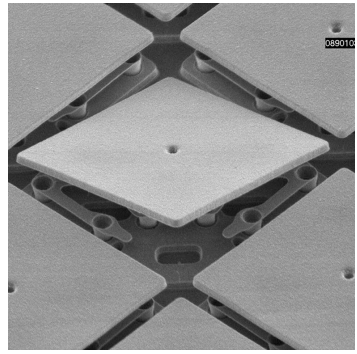
Optical Diagram



Digital Micromirror Device (DMD)



TI DLPDLCR471TPEVM evaluation module



Close-up of DMD micromirrors (IEEE Spectrum)[12]

DLP471TP DMD[13]: **5.4 µm mirror pitch**, 1920 × 1080 mirrors, ±17° tilt, 0.47” diagonal (10.37 × 5.83 mm)

With ≈10x demagnification: ≈0.54 µm pixel and ≈1.0 × 0.58 mm field on the wafer

Optical Specs

1. **Objective:** 10X DIN plan objective[14]
   1.1 NA = **0.25**
2. **Wavelength:** λ = **405 nm**
3. **Diffraction limit:**
   3.1 Airy disk diameter: 1.22 λ/NA = 1.98 µm
   3.2 Rayleigh resolution: 0.61 λ/NA = **0.99 µm**

What sets the 1.22?

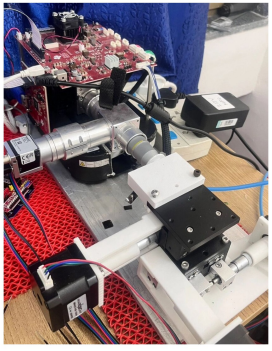
1. **Pupil shape:** 1.22 = 3.83/π, the first zero of the Bessel function J_1, for a uniformly lit *circular* aperture. A slit gives 1.0.
2. **Criterion:** Rayleigh 0.61, Abbe 0.5, Sparrow 0.47.
3. **Illumination coherence (σ):** coherent light resolves line/space down to a half-pitch of 0.5 λ/NA; off-axis illumination gets down to 0.25 λ/NA.
4. **In practice:** aberrations, defocus, the resist and the process push k_1 higher.

How we built Litho

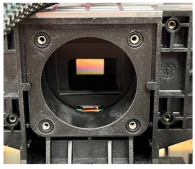
HOW WE BUILT LITHO: PROJECTOR TO TOOL

Five changes turn a DLP projector into a direct-write lithography tool

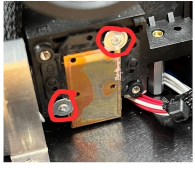
1. **Maskless:** the DMD projects any HDMI pattern, so no mask per design
2. **Demagnify:** unscrew the projection optics; mount a 10× objective in reverse
3. **UV:** replace the blue LED with a 405 nm LED board
4. **See the focus:** beamsplitter and camera image the wafer through the objective
5. **Move:** XYZ stage driven by three NEMA steppers; Z focuses, XY steps



Assembled tool



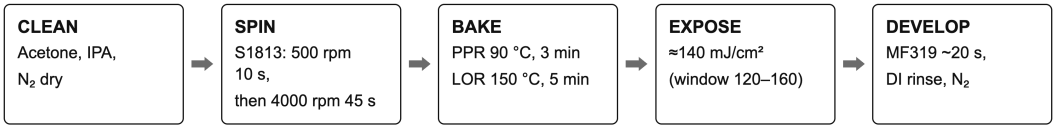
Optics removed



UV LED board

LITHO PROCESS: COAT, EXPOSE, DEVELOP

PPR alone for etching; LOR under PPR for sputter liftoff



- **LOR + PPR:** LOR is not photosensitive; developer undercuts it so sputtered metal lifts off cleanly
- **Dose matrix:** grid of exposure × development time finds the process window
- Window drifts with resist age, temperature and humidity; post-exposure bake not yet tested

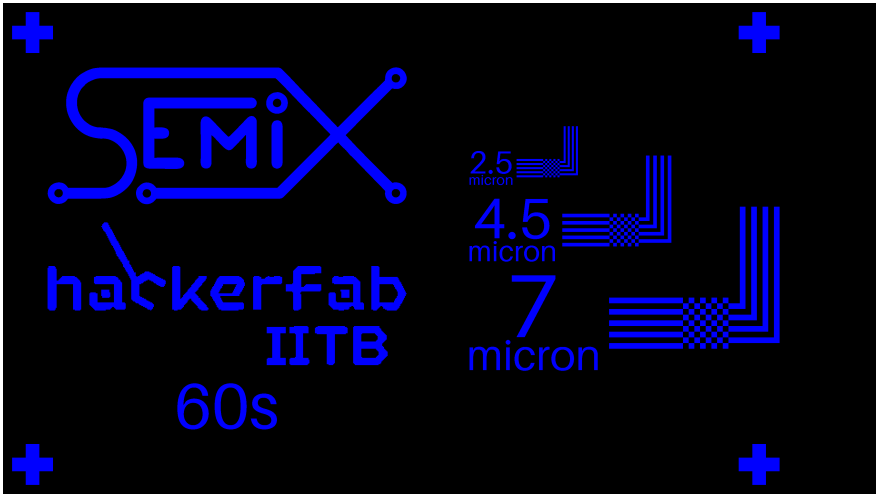
Q] Why only a 10x objective?

Why not use a 100x lens and get 500 nm resolution?

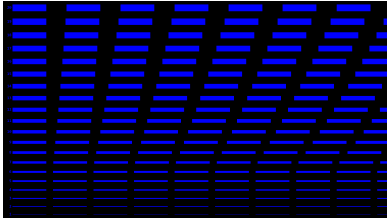
Characterising Performance of a Litho System

1. Critical Dimension (CD) / Minimum Feature Size / Resolution
2. Line/Space
3. Latent Image
4. Developed Image

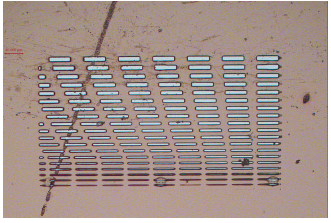
Logo Test Pattern



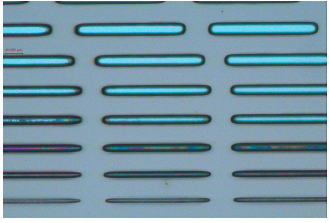
Resolution Test: 1–20 µm Lines



DMD pattern

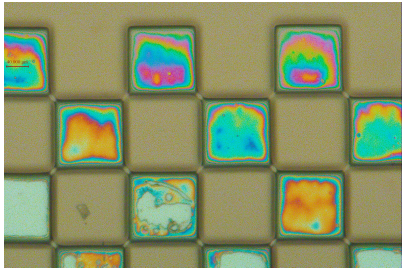


Exposed die, 10x

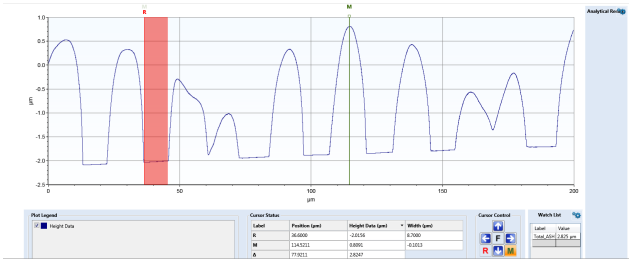


Exposed die, 50x

Interference Colors & Profilometer Scan

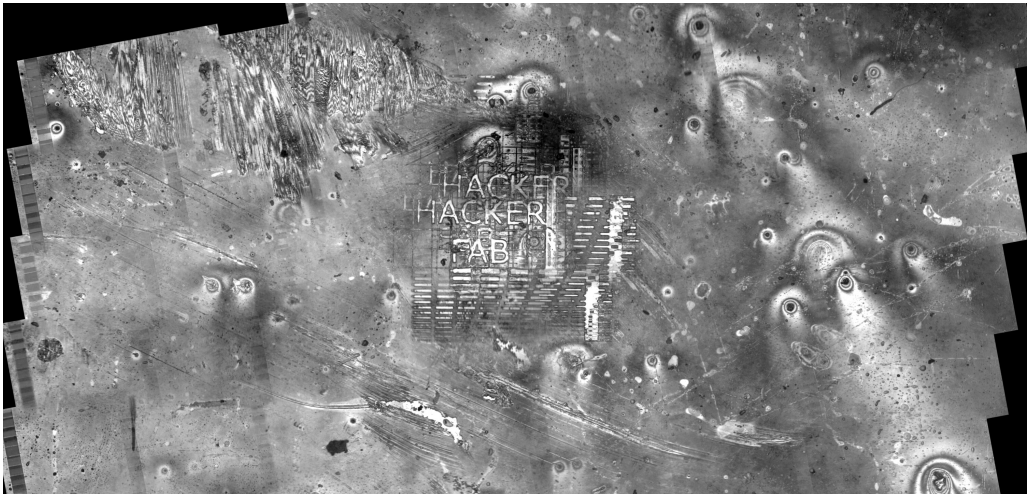


Interference colors, 100x

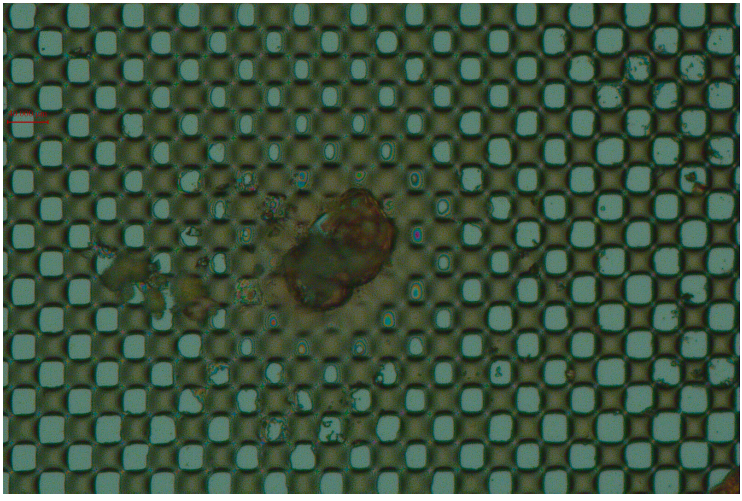


Profilometer scan: 10 µm squares

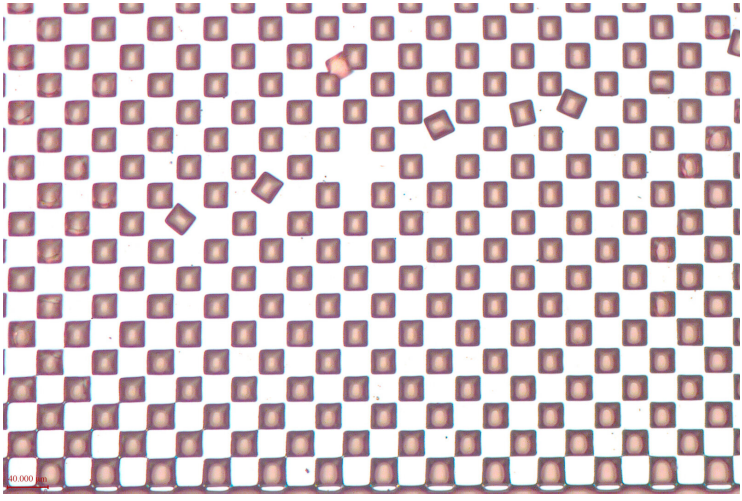
Wafer Fallen on the Floor



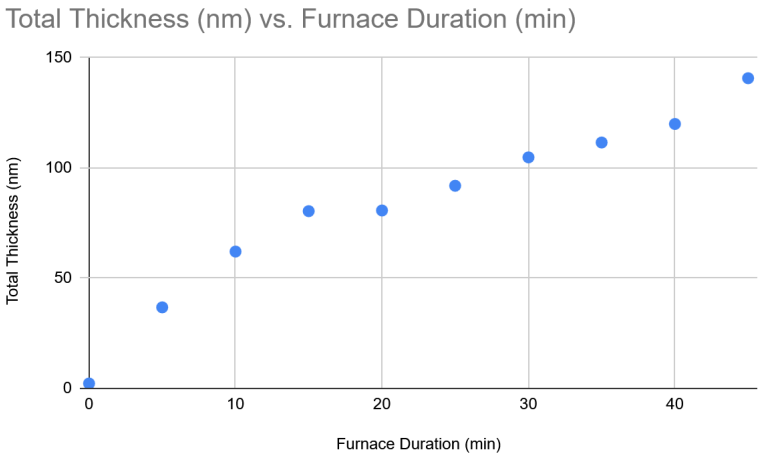
Dust Particle Defect



Moved Squares: Poor Adhesion



Furnace: Thickness vs. Duration

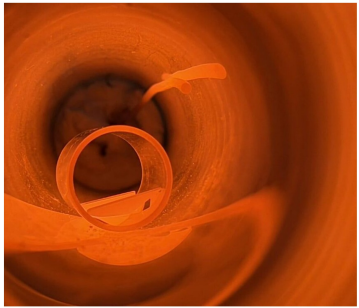


TUBE FURNACE: 1100 °C OXIDATION

Built in-house for about a third of a commercial furnace

Spec	Value
Reaction	$Si + O_2 \rightarrow SiO_2$
Temperature	1100 ± 5 °C, single zone
Tube	Quartz, 60 mm OD × 800 mm
Heater	Kanthal A1, 14 SWG coil
Controller	Multispan UTC-4202G+, K-type TC
Budget	₹35.5k (₹31.5k parts + ₹3.4k fab)

- Oxidation today; diffusion doping and other high-T steps next



Looking down the heated tube

FURNACE DESIGN: HEATER AND INSULATION

Coiled Kanthal cast in refractory cement, wrapped in ceramic wool

- Kanthal A1 14 SWG on an 8 mm mandrel, **3.5 mm pitch**; coil > 5 m
- Coil wound on a 75 mm PVC former at 3 cm spacing
- Cast in refractory cement inside 160 mm PVC; cure 3–4 days, watered
- PVC removed (grinder, torch); heat-cure ~½ day at 100 °C
- ~10 kg ceramic wool, grounded MS enclosure; quartz tube never touches cement



PVC mould



Coil on former



Cured cast

SPUTTER: SYSTEM OVERVIEW

DC magnetron sputter coater, mostly salvaged and locally made parts

ROTARY VANE PUMP
Roughs the line
atm → ~1e-3 mbar

→

TURBO PUMP
Runs continuously
→ 1e-5 mbar

→

CHAMBER
SS cylinder, 8–10 mm wall
Viewport, magnetron

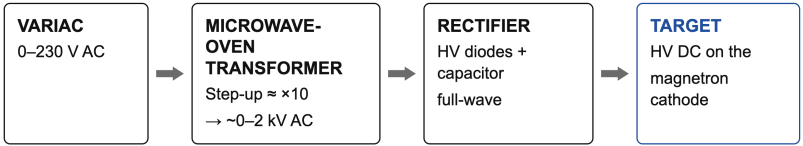
GAUGES
Pirani (KF16) to ~1e-3 mbar
Penning (KF25) below it, interlocked

GAS
Argon via MFC
strike ~0.1 mbar

- Base 1e-5 to 1e-6 mbar
- Sputter at 1e-2 to 1e-3 mbar
- 1 mbar: absolute max with turbo or Penning on
- No welds on sealing faces; PEEK screws isolate the top plate

MAGNETRON AND HIGH-VOLTAGE DC SOURCE

Ar^+ ions accelerated into the target knock metal atoms onto the wafer



MAGNETRON

- Magnets trap electrons near the target: denser plasma at lower pressure
- Mild steel body (it must be magnetic), electroless-nickel plated
- Magnet geometry follows HackerFab CMU

SAFETY

- Typical target bias −300 to −800 V
- Variac, capacitor and AC earth bonded to one grounded plate
- Microwave magnetron tubes may contain BeO: never scrape

FIRST FILMS: ALUMINIUM ON SILICON

Al sputtered at 115 W, thickness from profilometer steps

Run	Time	Step height	Observation
Sample 1	5 min	38–47 nm	Very dirty: black spots on Si and Al
Sample 2	15 min	not resolved	Cleaner; no flat reference beside the step

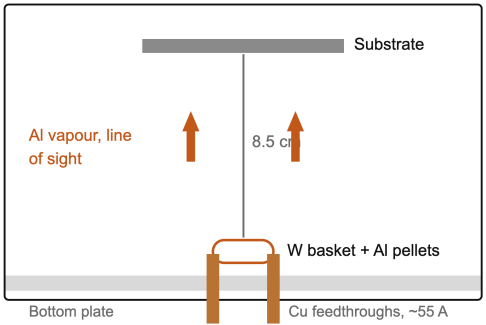
≈ 8–9 nm/min

deposition rate estimated from sample 1

- Contamination points to cleaning and base pressure; leaks are the usual cause
- Post-run step height is slow and can fail: we need in-situ thickness
- Lesson: an ISO-K to ISO-F adapter beats Viton on a CF knife-edge

THERMAL EVAPORATOR: SAME CHAMBER

Resistively heated tungsten basket in the sputter chamber



- **Reused:** chamber, rotary and turbo pumps, Pirani, Penning
- **Added:** two holes in the bottom plate; Cu feedthroughs, Teflon washers, Viton gaskets
- Welding PSU drives the basket: ramp 10 A every 30 s to **55 A**
- Evaporation starts after 6–7 min; run 10–12 min at 55 A

THE MISSING TURBO ISOLATION VALVE

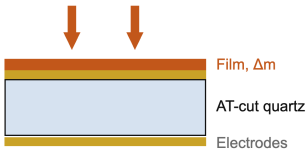
In evaporator mode, metal vapour has a straight path to the turbo blades

Problem	Consequence	Fix or plan
No gate valve between chamber and turbo	Hot metal vapour can reach and coat the blades: imbalance, mechanical failure	Gate valve or baffle in the next version
Chamber cannot hold vacuum on its own	Every run is a full pump-down: ~20 min rotary + ~30 min turbo	Isolation valve; load-lock (to-do)
Venting means stopping the turbo	Front off, ~15 min spin-down, back off; pressure creeps to 1e-4–1e-3 mbar	Spin-down in the SOP; cool filament 25 min before venting

- Shared chamber also cross-contaminates sputter and evaporation runs

QCM: QUARTZ CRYSTAL MICROBALANCE

Mass loading detunes a resonant crystal, so frequency becomes thickness



Oscillator → frequency counter: Δf

$$\Delta f = -2 f_0^2 \Delta m / (A \sqrt{(\rho_q \mu_q)})$$

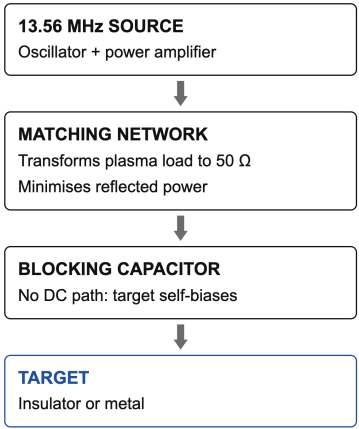
Sauerbrey relation, valid for thin rigid films.

- Crystal rings in thickness-shear at $f_0 \approx 5–6$ MHz
- 5 MHz crystal: ~18 ng/cm² per Hz ≈ **0.07 nm of Al per Hz**
- Thickness = Δm / (ρ_film · A), scaled by a tooling factor
- Gives live rate and end-point instead of post-run profilometry
- **Status:** our QCM is in development, not yet complete

WHY THE NEXT SPUTTER NEEDS RF

DC sputtering cannot run on an insulating target

- **DC:** Ar^+ charge builds up on an insulating target, the field collapses, the plasma dies or arcs
- **RF:** the field reverses every cycle, so surface charge is neutralised
- Electrons follow 13.56 MHz, heavy ions do not: the target self-biases **negative**, so ions still sputter it
- Opens up SiO_2, Al_2O_3 and other oxide or dielectric films
- Same supply still sputters metals



NEXT-VERSION SPUTTER: THE PLAN

Planned and in development: none of these is complete yet

Upgrade	Why
RF source (13.56 MHz) + matching network	Sputter insulators: SiO_2, Al_2O_3 and other oxide films
Turbo isolation gate valve or baffle	Protect the turbo; hold vacuum between runs
Separate evaporation chamber	No cross-contamination; independent operation
QCM thickness monitor	Live rate and thickness, repeatable depositions
Load-lock	Swap samples without venting the main chamber

Sources: sputter-v2 and thermal-evaporator-v1 pages, docs.hackerfabiitb.org

DOPING WITHOUT ION IMPLANTATION

Spin a dopant-loaded glass film, then drive it in with the furnace

- Industry: ion implantation, out of reach for us
- Commercial spin-on dopant (Filmtronics P504): ~₹1 lakh, corrosive, customs-handled
- **DIY SOD:** TEOS + water + catalyst in ethanol or IPA, plus dopant atoms
- Gel on a 175 °C hot plate: 20 min, then sample every 10 min until spinnable
- Anneal in the furnace: dopant diffuses from the glass into Si, $x_j \sim \sqrt{(D \cdot t)}$

WATCH-OUTS

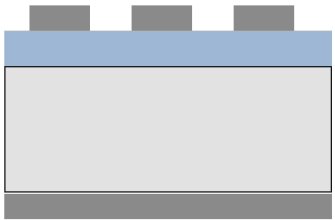
Viscosity depends on time, temperature and catalyst volume

Uneven film shows interference colours and dopes unevenly

Oxide masks doping: in NMOS, field oxide stays on so S/D remain visible

MOSCAP: TEST THE OXIDE FIRST

Six steps with shadow-mask metal; C–V is the readout



Not to scale

1. Clean the wafer piece
2. Oxidise at **1100 °C, 30 min**
3. Evaporate Al through a shadow mask: Al foil, pin-punched holes
4. Remove backside oxide: scratch, or HF with resist on the front
5. Optional: Al on the back for a contact
6. C–V on each dot (Proxima)

WHERE WE ARE, WHAT IS NEXT

A working toolchain; now the process work

Area	Status	Next
Litho	3 µm lines; 5/5 µm prints as 4/6 µm	Characterise stage, overlay, dose
Furnace	1100 ± 5 °C oxidation	Fit Deal–Grove to ellipsometry
Sputter	Al films, ~40 nm in 5 min	RF source, QCM (in development)
Evaporator	Al from a W basket at 55 A	Protect the turbo, own chamber
Processes	MOSCAP runs done; NMOS in progress	First working NMOS

- Everything is open-source: docs.hackerfabiitb.org

Thank you!

Questions?



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